DO NOW, AO1 REVISION

The development of the model of the atom

Atomic structure
Paper 1, Combined and Separate
14 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 5 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Name the first subatomic particle to be discovered.**	The electron	☐
2 ★	**Name the model of the atom with electrons embedded in a positively charged sphere.**	The plum pudding model	☐
3	**Name the model of the atom proposed after the nucleus was discovered.**	The nuclear model	☐
4	**Name the experiment that discovered the nucleus of an atom.**	The alpha particle scattering experiment	☐
5	**State what the alpha particle scattering experiment showed about the nucleus.**	The mass of the atom is concentrated in a very small, positively charged nucleus.	☐
6 ★	**Explain how the alpha particle scattering experiment showed that the nucleus is very small.**	Most alpha particles passed straight through; only a very few were scattered backwards.	☐
7 ★	**Explain how the alpha particle scattering experiment showed that the nucleus is positively charged.**	Some positive alpha particles were deflected or scattered backwards, repelled by the nucleus.	☐
8	**Name the model of the atom that came after the nuclear model.**	The Bohr model (Bohr adapted the nuclear model)	☐
9 ★	**Describe the Bohr model of the atom.**	A small positive nucleus with electrons orbiting at specific distances.	☐
10	**What were atoms thought to be before the electron was discovered?**	Tiny spheres that could not be divided.	☐
11	**Most alpha particles passed straight through the gold foil. State what this showed.**	Most of the atom is empty space.	☐

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	QUESTION	ANSWER	✓
12	**Explain why the alpha particle scattering experiment led to the plum pudding model being replaced.**	It could not explain why some alpha particles were deflected through large angles.	☐
13 ★	**Name the scientist who provided the evidence for the neutron.**	James Chadwick	☐
14	**Name the scientist who suggested that electrons orbit at specific distances.**	Niels Bohr	☐

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